

Benson (Zhen Cong) Chen

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AI Engineer specializing in low-level GPU optimization for machine learning systems across mobile, desktop, and safety-critical environments. Llama.cpp contributor with cross-vendor GPU experience and a track record of measurable wins—1.5× Ternary Qwen3 inference speedup on mobile GPUs, 60% runtime reduction on GEMM kernels, and BitNet ternary-quantization support across heterogeneous GPUs.

Skills & Competencies

- Programming languages: C, C++, Python, Java
- ML API: Vulkan, CUDA, Metal, PyTorch
- LLM: Llama.cpp, Safety Critical Environment, Extreme Quantization, GPU Kernel Optimization, AI Inference, Edge Deployment

Work Experience

AI Research Engineer (Edge Inference) **Tether.io** **JAN 2026 – PRESENT**

- Develop and optimize kernels, quantization workflows, and fine-tuning systems for LLMs across mobile and desktop GPUs.
- Optimized Qwen3 model on mobile Adreno 840 GPU to achieve 1.5× speedup compared to the previous version.
- Added support for BitNet heterogeneous GPU inference in llama.cpp by enabling ternary quantization with GPU acceleration using Vulkan and Metal.
- Authored a research paper documenting the implementation of BitNet LoRA fine-tuning on heterogeneous GPUs.
- Research and benchmark current SOTA LLM inference technique to integrate into the company's edge inference system.

ML Software Developer (ML Inference) **Lynx Software** **JAN 2024 – DEC 2025**

- Benchmarked and optimized GEMM operations on an Imagination GPU, reducing application runtime by 60% through memory tiling, register tiling, and thread coarsening; validated bottleneck-level gains using a custom profiling tool.
- Enabled efficient linear quantization for matrix multiplication to improve computational performance.
- Optimized GPU computation for safety-critical machine learning operations such as convolution.
- Implemented a testing library for the ML inference engine to thoroughly test each ML operation for various inputs using PyTorch.
- Designed and built a data processing pipeline for an internal LLM tool.
- Automated software testing by leveraging Jenkins and Python scripts.

ML Software Developer (ML Inference) **CoreAVI (Lynx Software)** **MAY 2023 – AUG 2023**

- Implemented various machine learning operations using Vulkan compute shaders conforming to the NNEF specification into company's custom machine learning engine.
- Designed debugging tools for developers working on the machine learning engine.
- Built demo projects running VGG16 and VGG18 vision models on the company's machine learning framework to perform object detection on Intel and AMD GPUs.

ML Software Developer (ML Inference) **CoreAVI (Lynx Software)** **SEPT 2022 – DEC 2022**

- Enhanced the machine learning engine to support additional ML architectures.
- Refined the NNEF compiler to parse NNEF documents according to the NNEF specification.

Compiler Software Engineer **Huawei** **SEPT 2021 – DEC 2021**

- Generated and preprocessed training data for the AI compiler used to predict the best optimization for LLVM IR.
- Created an API for an AI compiler ML model to perform predictions on LLVM IR using SSH and network sockets.

Education

The University of Texas at Austin Master of Computer Science, Candidate Present

University of Waterloo Bachelor of Computer Science (Co-operative Program), Dean's Honour 2023